

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 02
Duration: 20 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name: Solution	ID:	Section:06
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1. Why do we use stack in Shift-Reduce Parsing? (2.5)
2. What are the conflicts in Shift-Reduce Parsing? (2.5)
3. Show the Handles during the parsing of $id1 + id2 + id3$ (10)

Right Sentential Form	Handle	Reducing Productions
$id1 + id2 + id3$		$T \rightarrow id$
$T + id2 - id3$		$E \rightarrow T$
$E + id2 - id3$		$T \rightarrow id$
$E + T - id3$		$E \rightarrow E + T$
$E - id3$		$T \rightarrow id$
$E - T$		$E \rightarrow E - T$

Question 01:

The use of a stack in shift-reduce parsing is justified by an important fact:

1. the handle will always eventually appear on top of the stack, never inside.
2. It uses Last in First Out technique.

Question 02:

There are context-free grammars for which shift-reduce parsing cannot be used. Every shift-reduce parser for such a grammar can reach a configuration in which the parser, knowing the entire stack contents and the next input symbol, cannot decide whether to shift or to reduce (a shift/reduce conflict), or cannot decide which of several reductions to make (a reduce/reduce conflict).

Question 03:

Right Sentential Form	Handle	Reducing Productions
$id1 + id2 + id3$	$id1$	$T \rightarrow id$
$T + id2 - id3$	T	$E \rightarrow T$
$E + id2 - id3$	$id2$	$T \rightarrow id$

$E + T - id_3$	$E + T$	$E \rightarrow E + T$
$E - id_3$	id_3	$T \rightarrow id$
$E - T$	$E - T$	$E \rightarrow E - T$